

SCHOOL OF COMPUTING (SEM II – 2021/2022)

SECI2143 – PSDA

Exercise (Chapter 4)

QUESTION 1

- a) On 17th March 2020, Malaysia recorded its first death of COVID-19. Since then, the number of death is increasing. Table 7 shows the probability distribution of number of deaths in a day for 22 days' period since 17th March 2020.

Table 7: Probability distribution of number of deaths is a day

0	1	2	3	4	5	6	7	8
0.091	0.182	0.182	0.227	0.137	0.091	0.045	0.000	0.045

- i. Find the probability that at most 2 deaths occur in a day. (1 mark)
 - ii. Find the probability that at least 5 deaths occur in a day. (1 mark)
 - iii. Find the probability that at least 3 deaths but not more than 5 deaths occur in a day. (1 mark)
 - iv. Based on Table 7 data for 22 days, find mean and variance of the number of deaths occur in a day. (2 marks)
- b) According to World Health Organization (WHO), the risk of dying if infected by COVID-19 for a patient with a given pre-existing condition is shown in Table 8.

Table 8: Probability of dying for a patient with a given pre-existing condition

Pre-existing Condition	Probability
Cardiovascular Disease	0.105
Diabetes	0.073
Chronic Respiratory Disease	0.063
Hypertension	0.060
Cancer	0.056
No Pre-existing Conditions	0.009

Suppose that among 56 COVID-10 patients in a hospital, there are 8 patients with cardiovascular disease, 4 patients with diabetes, 10 patients with chronic respiratory disease, 8 patients with hypertension and 5 patients with cancer.

- i. Suppose that patients with cardiovascular disease are selected at random, what is the probability that at most two patients will die? (2 marks)
- ii. Suppose that patients with diabetes are selected at random, what is the probability that at least two patients will die? (2 marks)
- iii. Suppose that patients with chronic respiratory disease are selected at random, what is the probability that at most one patients will recover? (1½ marks)
- iv. Suppose that patients with cancer are selected at random, what is the probability that at least two but no more than four patients will recover? (1½ marks)
- v. Suppose that patients with no pre-existing conditions are selected at random, what is the probability that all patients will recover? (1 mark)

QUESTION 2

In the recent COVID-19 outbreak, 58% of the suspected cases who goes for screening test get positive result. It is known that 28% of the confirmed positive cases has history of traveling to the country with outbreak, 50% has close contact with confirmed positive person and another 22% is from unknown cluster.

- a) Represent the probability of the COVID-19 outbreak described above using tree diagram.
You can see examples of tree diagram in <https://www.mathsisfun.com/data/probability-tree-diagrams.html> (3 marks)
- b) What is the probability of suspected cases with negative screening test result? (1 mark)
- c) If a total of 72 person-under-investigation (PUI) was suspected to be exposed to this virus and are required to take swab test. How many people will get negative result? (1 mark)
- d) Find the probability of positive cases having travelling history to the country with outbreak or close contact with person infected? (2 marks)
- e) Death rate of this pandemic is 1.6% while the recovery rate is 34%. If the total infected person is 2766 people,
 - i. What is the probability of infected person under treatment? (2 marks)
 - ii. How many are under treatment? (1 mark)

QUESTION 3

- a) An agricultural corporation claims that 90% of the pineapples shipped out are ripe and ready to eat. Find the probability that among 18 pineapples shipped out, all 18 pineapples are ripe and ready to eat.
- b) A tourism transportation company in a small town runs a fleet of 20 vans, which are all aging. Assume that the probability of some type of maintenance being required, for each van each day is 0.2, and that they will or will not require maintenance independently. If Y is the number of vans that require maintenance on a given day:
 - i. What is the mean for Y ?
 - ii. What is the variance for Y ?
- c) An expert shot hits a target 95% of the time. What is the probability that the expert will miss the target for the first time on the fifteenth shot?

QUESTION 4

Table 2 below shows the number of hospitalizations for cardiovascular disease, by age group, for one year.

Age group (year)	Number
0-19	60
20-39	150
40-49	300
50-59	580
60-69	678
70-79	1288
80 and over	1378

One of these case records is selected at random. Find the probability that the selected patient was

- a) In the age of 50s
- b) Less than 50 years old
- c) Between 40 and 69 years old, inclusive.
- d) At least 70 years old.

QUESTION 5

In a Japanese restaurant, an average of 3 out of every 5 customers ask for plain water with their meal. A random sample of 10 customers is selected. Let X equals to the number of customers that ask for plain water. Assume each trial is independent.

- a) Find the probability that exactly 6 ask for plain water with their meal.
- b) What is the probability that less than 9 customers ask for plain water with their meal?
- c) A second random sample of customers is selected. What is the probability that the first who to ask for plain water is the fourth customer?

QUESTION 6

- a) 60% of all computers sold by a large computer retailer are laptops and 40% are desktop models. The type of computer purchased by each of the next 12 customers will be recorded. Given a random variable x as number of laptops purchased among these 12 customers, determine
- Either x is a binomial random variable or geometry random variable.
 - The probability that exactly four computers are laptops.
 - The probability that between four and seven (inclusive) are laptops.
 - Given a random variable x as the number of desktops purchased among these 12 customers, determine the probability that exactly two computers are desktops.
- b) From the experience it is known that 3% of accounts in a large company accounting population are in error.
- What is the random variable x representing?
 - What is the probability that 5 accounts ($x=5$) are audited before an account in error is found?
 - What is the probability that the first account in error occurs is the first five accounts ($x \leq 5$) audited?